

LITERATURE REVIEW

Smart Waste Management Using Big Data Analytics for
Toxicity-Aware Overflow Prediction and Dynamic Collection Route
Optimization

Submitted by

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1. Introduction

Smart waste management is increasingly supported by IoT, machine learning, Big Data analytics and optimization techniques. Traditional waste collection often uses fixed schedules and predefined routes, which may not reflect actual waste accumulation, overflow risk, waste toxicity or differences between industrial and rural areas. Recent research has therefore explored smart monitoring, prediction, real-time decision-making and vehicle route optimization.

This literature review examines recent research related to IoT-enabled waste monitoring, overflow prediction, machine-learning-based waste management, vehicle routing and real-time collection decisions. The reviewed studies provide the foundation for a proposed Big Data workflow using HDFS, YARN and Apache Spark.

2. Objectives of the Literature Review

- Identify recent approaches for smart waste monitoring and collection.
- Study the use of IoT, machine learning and optimization in waste management.
- Examine methods for route optimization and cost reduction.
- Identify research gaps related to toxicity, overflow, area classification and distributed Big Data processing.

3. Literature Review

3.1 IoT-Assisted Vehicle Route Optimization for Municipal Solid Waste Collection

Year: 2024

Authors: Bakhtiar Ali, Muhammad Awais Javed, Abeer A. K. Alharbi, Salihah Alotaibi, Mohammed Alkhathami

Focus and Contribution: The study presents an IoT-assisted approach for monitoring municipal solid waste and improving vehicle routes. Waste information is used to support more efficient collection planning and reduce unnecessary travel.

Limitation / Gap: The work mainly concentrates on IoT monitoring and route optimization. It does not integrate toxicity-based priority, area-wise classification, overflow prediction, and Hadoop/Spark-based large-scale processing as one workflow.

Paper Link: <https://www.mdpi.com/2076-3417/14/1/287>

3.2 Multi-objective Location-Routing Optimization Based on Machine Learning for Green Municipal Waste Management

Year: 2024

Authors: Yunyun Niu, Chang Xu, Shubing Liao, Shuai Zhang, Jianhua Xiao

Focus and Contribution: This research combines machine learning with multi-objective location-routing optimization for municipal waste management. It considers waste demand, routing

distance, total cost and environmental objectives.

Limitation / Gap: The focus is on location-routing and optimization. Toxicity-aware collection priority and explicit overflow prediction with a Hadoop/Spark Big Data pipeline are not combined in the reviewed approach.

Paper Link: <https://www.sciencedirect.com/science/article/pii/S0956053X24001910>

3.3 Smart Waste Management System Using IoT

Year: 2024 (published 2025)

Authors: IEEE Silchar Subsection Conference (SILCON 2024)

Focus and Contribution: The paper discusses an IoT-based smart waste management system involving fill-level and environmental monitoring, cloud processing, AI-based analysis, prediction and disposal scheduling.

Limitation / Gap: The reviewed work emphasizes smart monitoring and prediction. It does not provide the complete combination of toxicity-based prioritization, industrial/rural classification, overflow risk and dynamic cost-aware routing using Hadoop and Spark.

Paper Link: <https://ieeexplore.ieee.org/document/10910604>

3.4 Vehicle Route Optimizer for Waste Collection and Routing Optimization Problem

Year: 2025

Authors: Hussam Fakhouri, Amjad Hudaib, Faten Hamad, Sandi Fakhouri, Niveen Halalsheh, Mohannad S. Alkhalaileh

Focus and Contribution: The study proposes a vehicle route optimizer for waste collection and routing. It addresses bin allocation and vehicle routing with attention to total operational cost and computational efficiency.

Limitation / Gap: The main contribution is routing optimization. The reviewed approach does not combine route optimization with toxicity classification, overflow prediction and distributed Hadoop/Spark analytics.

Paper Link: <https://www.sciencedirect.com/science/article/pii/S266730532500047X>

3.5 Real-Time Operational Decision Making in Municipal Waste Collection Systems Using IoT Technologies

Year: 2025

Authors: Tamás Bányai, Sajid Nazir, Péter Veres

Focus and Contribution: This work investigates real-time decision-making in municipal waste collection using IoT, AI, Big Data analytics and cloud technologies. It considers monitoring, predictive analytics, dynamic routing, scheduling and resource allocation.

Limitation / Gap: Although it addresses real-time operational decisions, the reviewed work does not specifically combine toxicity-aware priority, industrial/rural area classification, overflow prediction and Hadoop/Spark implementation in the proposed project structure.

Paper Link: <https://www.sciencedirect.com/science/article/pii/S2405896325007888>

4. Comparative Analysis

Study	IoT / Monitoring	Prediction / ML	Route Optimization	Toxicity + Overflow	Hadoop / Spark
Ali et al. (2024)	Yes	Limited	Yes	Not combined	No
Niu et al. (2024)	Not primary	Yes	Yes	Not combined	No
SILCON (2024/2025)	Yes	Yes	Scheduling	Not combined	No
Fakhouri et al. (2025)	Limited	Limited	Yes	Not combined	No
Bányai et al. (2025)	Yes	Yes	Yes	Not combined	Big Data / cloud
Proposed approach	Data source	Overflow prediction	Dynamic	Yes	HDFS + YARN + Spark

5. Research Gap

The reviewed studies show that different aspects of smart waste management have been addressed separately or in partial combinations. IoT-based monitoring can provide information about waste conditions, machine learning can support prediction, and optimization algorithms can improve collection routes. However, the reviewed papers do not combine all of the following dimensions in the same project workflow: **area classification, toxicity-based priority, overflow prediction, dynamic collection routing and distributed Hadoop/Spark processing.**

Therefore, there is an opportunity to develop an integrated Big Data analytics framework that processes large-scale waste records, identifies collection urgency and generates dynamic routes while considering transportation distance, fuel, time and operational cost.

6. Existing System

- Fixed collection schedules and predefined routes.
- Limited use of real-time or historical fill-level information.
- Bins may be treated similarly even when waste toxicity differs.
- Industrial and rural waste characteristics may not be explicitly differentiated.
- Overflow can occur before the scheduled collection.
- Unnecessary travel can increase fuel usage, collection time and transportation cost.

7. Proposed System

The proposed system is a software-based Big Data analytics solution. Waste records can be obtained from IoT-style data sources, public datasets or simulated sensor data; physical sensors are not mandatory for the project implementation. Large-scale records are stored in HDFS and managed

through YARN. Apache Spark is used for distributed preprocessing, analysis and prediction.

Proposed processing stages:

1. Collect large-scale waste data.
2. Store the data in HDFS.
3. Use YARN for distributed resource management.
4. Process the data using Apache Spark / PySpark.
5. Classify locations such as industrial, rural, urban and commercial areas.
6. Analyse toxicity level, fill level, accumulation rate and historical collection patterns.
7. Predict overflow risk using machine learning.
8. Calculate collection priority using toxicity and urgency factors.
9. Generate dynamic collection routes considering priority, distance, vehicle capacity and travel time.
10. Compare routes and estimate distance, fuel, time and transportation-cost savings.

8. Proposed Big Data Architecture

Waste Data Sources → HDFS → YARN → Apache Spark → Data Preprocessing → Area Classification → Toxicity Analysis + Overflow Prediction → Collection Priority → Dynamic Route Optimization → Distance/Fuel/Time/Cost Analysis → Dashboard

9. Technologies Considered

- **HDFS:** Distributed storage for large-scale waste data.
- **YARN:** Resource management for the Hadoop cluster.
- **Apache Spark:** Distributed data processing and analytics.
- **PySpark:** Python-based implementation of Spark processing.
- **Spark MLlib:** Machine-learning support for overflow prediction.
- **Routing algorithm:** Dynamic route generation based on collection priority and operational constraints.
- **Dashboard:** Visualization of overflow risk, priority bins, routes and performance metrics.

10. Expected Outcomes

- Early identification of bins with high overflow risk.
- Higher collection priority for high-toxicity waste.
- Area-aware waste collection decisions.
- Dynamic rather than purely fixed collection routes.
- Reduction in unnecessary travel distance, fuel usage and collection time.

- Data-driven estimation of transportation and operational cost.
- Scalable processing of large waste datasets using Hadoop and Spark.

11. Conclusion

Recent literature demonstrates significant progress in IoT-based waste monitoring, machine learning, real-time collection decisions and route optimization. The reviewed studies provide useful methods for individual parts of the proposed problem. The proposed work brings these ideas into an integrated Big Data workflow with particular attention to toxicity, overflow risk, area classification and dynamic routing. HDFS, YARN and Apache Spark provide the distributed processing foundation for handling large-scale waste records and supporting data-driven collection decisions.

12. References

1. Bakhtiar Ali, Muhammad Awais Javed, Abeer A. K. Alharbi, Salihah Alotaibi, Mohammed Alkhathami. *IoT-Assisted Vehicle Route Optimization for Municipal Solid Waste Collection*. 2024. <https://www.mdpi.com/2076-3417/14/1/287>
2. Yunyun Niu, Chang Xu, Shubing Liao, Shuai Zhang, Jianhua Xiao. *Multi-objective Location-Routing Optimization Based on Machine Learning for Green Municipal Waste Management*. 2024. <https://www.sciencedirect.com/science/article/pii/S0956053X24001910>
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4. Hussam Fakhouri, Amjad Hudaib, Faten Hamad, Sandi Fakhouri, Niveen Halalsheh, Mohannad S. Alkhalaileh. *Vehicle Route Optimizer for Waste Collection and Routing Optimization Problem*. 2025. <https://www.sciencedirect.com/science/article/pii/S266730532500047X>
5. Tamás Bányai, Sajid Nazir, Péter Veres. *Real-Time Operational Decision Making in Municipal Waste Collection Systems Using IoT Technologies*. 2025. <https://www.sciencedirect.com/science/article/pii/S2405896325007888>